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SYMBIOSIS INSTITUTE OF TECHNOLOGY, PUNE

Department of Artificial Intelligence and Machine Learning (AIML)

Assignment Practical No. 6

Adventure Works Sales & Profitability Dashboard

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Course	Data Visualization Lab (TE7663)
Semester / Year	Sem V, Academic Year 2026-2027

PUBLISH LINK

https://github.com/ManavG27/DATA-VIZUALIZATION-BLOG/blob/main/Assignment_Practical_6_Manav_Goel-3.pdf

1. Aim and Objectives

This practical aims to design analytical Key Performance Indicators (KPIs) and custom DAX calculations in Power BI Desktop, to validate them using DAX Studio, and to present the results through an interactive dashboard. The analysis addresses the following question: which products, categories and regions contribute most to the sales and profit of Adventure Works between January 2020 and June 2022, and how does performance change over time?

The objectives follow the six tasks specified for this practical. Each task is addressed in a dedicated section of this report, as shown below.

Task	Objective	Section
1	To create the required KPI measures: Total Sales, Total Cost, Total Profit, Total Quantity, Total Orders, Total Customers, Average Sales, Average Order Value, Average Product Price and Profit Margin %.	Section 3
2	To implement at least twenty custom DAX measures using functions such as SUM, AVERAGE, COUNT, DISTINCTCOUNT, CALCULATE, DIVIDE, FILTER and RANKX.	Section 4
3	To perform time-based analysis covering monthly sales, previous-period sales, sales variance and sales growth percentage.	Section 5
4	To execute DAX queries in DAX Studio, validate the measures and record the results.	Section 6
5	To develop an interactive dashboard with KPI cards, at least four visualizations, and slicers for year, product category and region.	Section 7
6	To analyze the results by identifying top-selling products, highest-performing categories and regions, sales trends and profitability, and to state the key insights and conclusions.	Sections 8 and 9

Expected deliverables. DAX measures and KPI calculations (Sections 3 to 5), DAX Studio queries and screenshots (Section 6), the interactive Power BI dashboard (Section 7), and the analytical findings and conclusion (Sections 8 and 9).

2. Dataset Description and Data Preparation

The Adventure Works dataset was used, and only the tables required for the analysis were loaded into the model. The data spans 1 January 2020 to 30 June 2022; consequently, the year 2022 represents a half-year period.

Table	Rows	What it holds
Sales (2020–2022 combined)	56,046	Every order line: date, order number, product, customer, territory, quantity
Products	293	Product names, costs and prices
Subcategories	37	Connects products to categories
Categories	4	Bikes, Components, Clothing, Accessories
Territories	10	Region, country and continent
Customers	18,148	Customer details (after cleaning)
Dates	912	Calendar from 1 Jan 2020 to 30 Jun 2022

Data cleaning. The three yearly sales files were combined into a single Sales table using the Folder connector in Power Query. Six non-customer records were removed from the Customers file (three rows with a key of the form “30---”, one blank row and two export-note lines), and the CustomerKey column was converted to the Whole Number data type. This resolved the duplicate-null error and enabled the relationship between the Sales and Customers tables.

Calculated columns. The sales files contain order quantities only. Sales and cost amounts were therefore derived by multiplying the quantity by the price and cost of the corresponding product:

```
Sales Amount = Sales[OrderQuantity] * RELATED(Products[ProductPrice])
Cost Amount  = Sales[OrderQuantity] * RELATED(Products[ProductCost])
```

Data model. All relationships in the model are many-to-one with a single filter direction: Sales to Products, Customers, Territories and Dates, and Products to Subcategories to Categories. The Dates table contains the Year, Month Number, Month Name and YearMonth columns and is designated as the date table so that time-intelligence functions operate correctly.

3. KPI Measures (Task 1)

The ten required KPI measures were created in a dedicated Measures table. They are listed below in the order specified in the practical, together with the requirement that each one satisfies.

Sr. No.	Measure	DAX	Requirement
1	**Total Sales**	`SUM(Sales[Sales Amount])`	Calculate total sales amount
2	**Total Cost**	`SUM(Sales[Cost Amount])`	Calculate total product cost
3	**Total Profit**	`[Total Sales] - [Total Cost]`	Total Sales − Total Cost
4	**Total Quantity**	`SUM(Sales[OrderQuantity])`	Calculate total order quantity
5	**Total Orders**	`DISTINCTCOUNT(Sales[OrderNumber])`	Count total orders
6	**Total Customers**	`DISTINCTCOUNT(Sales[CustomerKey])`	Count unique customers
7	**Average Sales**	`AVERAGE(Sales[Sales Amount])`	Calculate average sales
8	**Average Order Value**	`DIVIDE([Total Sales], [Total Orders])`	Total Sales ÷ Total Orders
9	**Average Product Price**	`AVERAGE(Products[ProductPrice])`	Average product price
10	**Profit Margin %**	`DIVIDE([Total Profit], [Total Sales]) * 100`	Profit ÷ Sales × 100

DIVIDE is used in the ratio measures so that a zero denominator returns a blank instead of an error. Profit Margin % is multiplied by 100 so that it is expressed as a percentage, as the practical specifies.

4. Custom DAX Calculations (Task 2)

Eleven further measures were created in addition to the ten KPIs. Together with the four time-based measures in Section 5, the model holds twenty-five measures, above the required minimum of twenty.

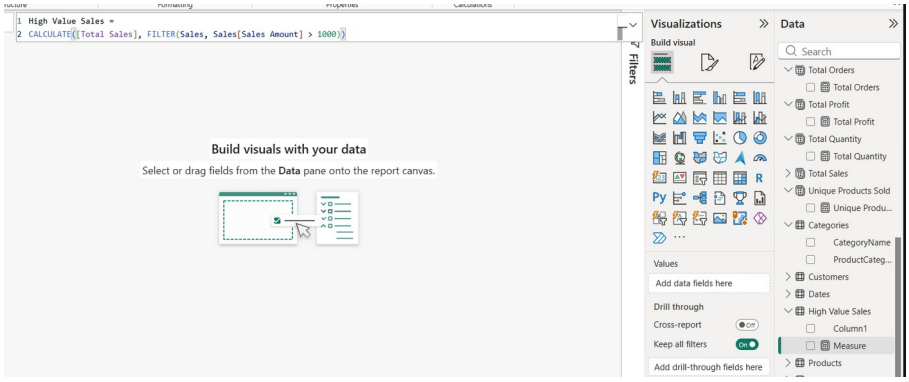
Sr. No.	Measure	DAX	What it gives
11	**Order Line Items**	`COUNT(Sales[OrderLineItem])`	Number of order lines
12	**Sales per Customer**	`DIVIDE([Total Sales], [Total Customers])`	Revenue per customer
13	**Profit per Order**	`DIVIDE([Total Profit], [Total Orders])`	Profit per order
14	**Avg Quantity per Order**	`DIVIDE([Total Quantity], [Total Orders])`	Units per order
15	**Unique Products Sold**	`DISTINCTCOUNT(Sales[ProductKey])`	How many products sold
16	**Bikes Sales**	`CALCULATE([Total Sales], Categories[CategoryName] = "Bikes")`	Sales from Bikes only
17	**High Value Sales**	`CALCULATE([Total Sales], FILTER(Sales, Sales[Sales Amount] > 1000))`	Sales from lines over 1,000
18	**Sales % of Total**	`DIVIDE([Total Sales], CALCULATE([Total Sales], ALLSELECTED())) * 100`	Share of selected total
19	**Product Rank**	`RANKX(ALL(Products[ProductName]), [Total Sales], , DESC, DENSE)`	Product ranking
20	**Region Rank**	`RANKX(ALL(Territories[Region]), [Total Sales], , DESC, DENSE)`	Region ranking
21	**Category Rank**	`RANKX(ALL(Categories[CategoryName]), [Total Sales], , DESC, DENSE)`	Category ranking

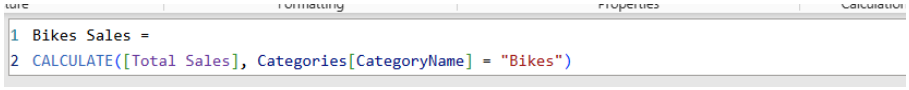
Functions Covered

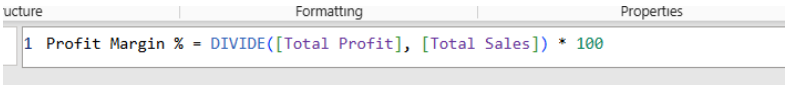
Every function named in the practical is used in at least one measure:

Function	Used in
SUM	Total Sales, Total Cost, Total Quantity
AVERAGE	Average Sales, Average Product Price
COUNT	Order Line Items
DISTINCTCOUNT	Total Orders, Total Customers, Unique Products Sold
CALCULATE	Bikes Sales, High Value Sales, Sales % of Total, Monthly Sales, Previous Month Sales
DIVIDE	Average Order Value, Profit Margin %, Sales per Customer, Profit per Order, Avg Quantity per Order, Sales % of Total, Sales Growth %
FILTER	High Value Sales
RANKX	Product Rank, Region Rank, Category Rank

Illustrative Screenshots of DAX Measure Creation







5. Time-Based Analysis (Task 3)

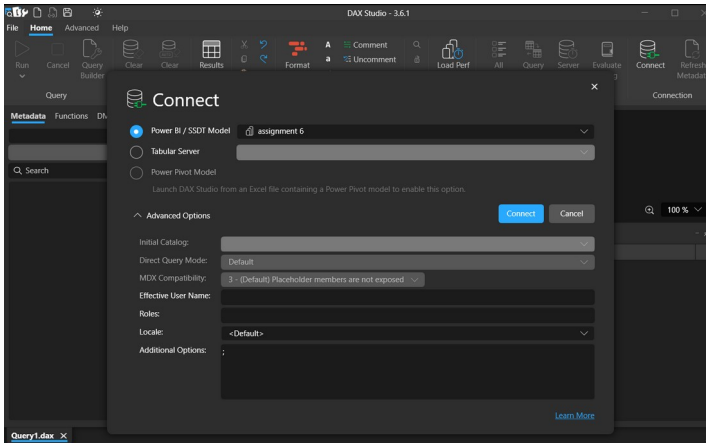
Four measures were created to analyze sales over time. They cover monthly sales, previous-period sales, sales variance and sales growth percentage.

Sr. No.	Measure	DAX	What it gives
22	**Monthly Sales**	`CALCULATE([Total Sales], DATESMTD(Dates[Date]))`	Sales for the month
23	**Previous Month Sales**	`CALCULATE([Total Sales], DATEADD(Dates[Date], -1, MONTH))`	Last month's sales
24	**Sales Variance**	`[Total Sales] - [Previous Month Sales]`	Change from last month
25	**Sales Growth %**	`DIVIDE([Sales Variance], [Previous Month Sales]) * 100`	Month-on-month growth

Sales Variance is the difference between the current and the previous month, and Sales Growth % expresses that difference as a percentage of the previous month. Because DIVIDE is used, the first month in the data, which has no previous month, returns a blank growth value. The month-by-month output of these measures is shown in Query 5 in Section 6.

6. Query Execution and Validation in DAX Studio (Task 4)

Because the External Tools tab was not available in the Power BI ribbon, DAX Studio was launched independently and connected to the open Power BI file through the PBI / SSDT Model option. Queries could then be executed directly against the data model.



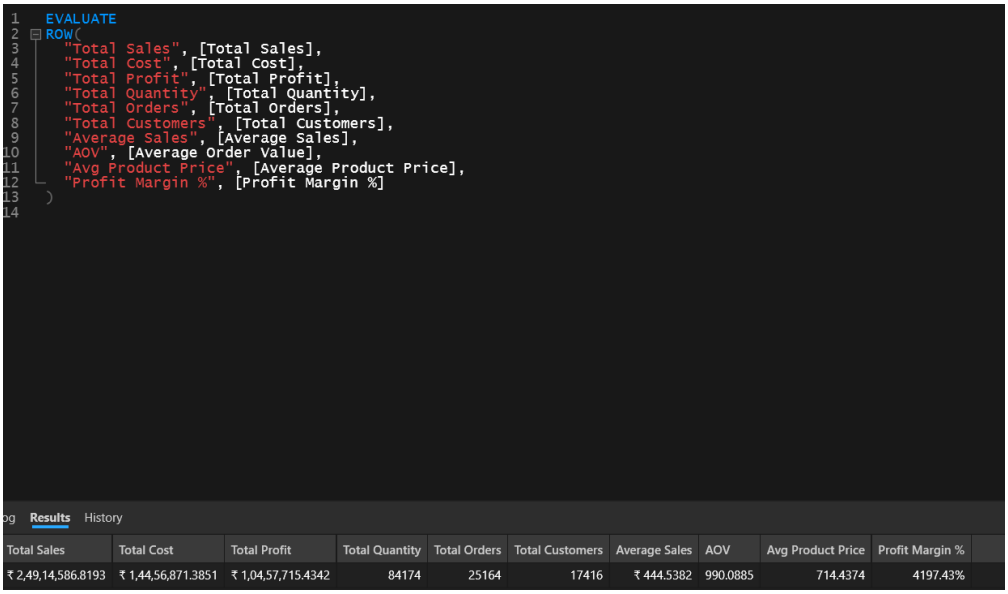
Seven queries were executed in DAX Studio. Each addresses a specific part of the validation and analysis, as summarized below.

Query	Purpose	Relates to
Query 1	Returns the ten required KPIs in one row for comparison with the dashboard cards	Task 1: validate KPIs
Query 2	Ranks the top ten products by sales (first rows shown)	Task 6: top-selling products
Query 3	Sales and profit margin by category and region	Task 6: categories and regions
Query 4	Complete top-ten product ranking with sales and profit	Task 6: top-selling products
Query 5	Monthly sales, previous month, variance and growth %	Task 3: time-based analysis
Query 6	Query-scoped test measure compared with Profit Margin %	Task 1: validate KPIs
Query 7	Lists every measure and its DAX expression stored in the model	Task 2: measure audit

Query 1: Validation of KPI Measures

This query returns all required KPI measures in a single row so that they can be compared with the values displayed on the dashboard cards.

```
EVALUATE
ROW(
  "Total Sales", [Total Sales],
  "Total Cost", [Total Cost],
  "Total Profit", [Total Profit],
  "Total Quantity", [Total Quantity],
  "Total Orders", [Total Orders],
  "Total Customers", [Total Customers],
  "Average Sales", [Average Sales],
  "AOV", [Average Order Value],
  "Avg Product Price", [Average Product Price],
  "Profit Margin %", [Profit Margin %]
)
```



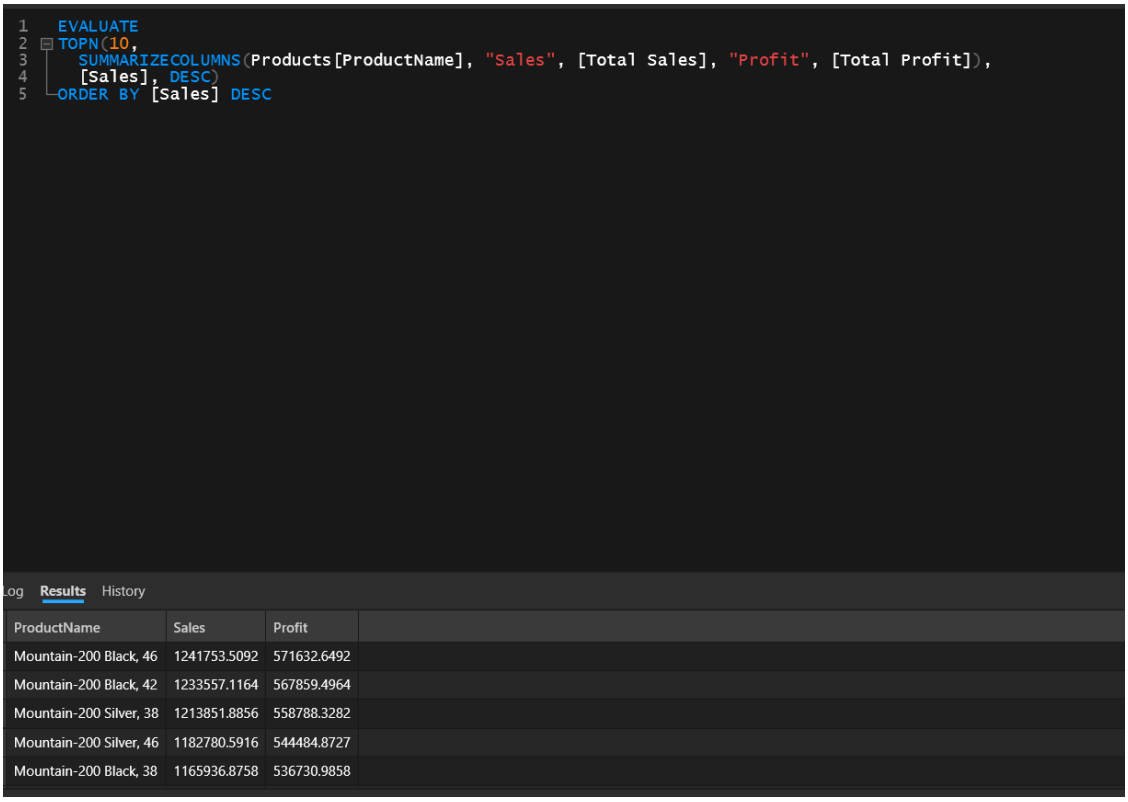
The same totals were also computed directly from the raw CSV files, independently of Power BI. All values agreed with the model output:

Measure	Value	Matches Raw Data?
Total Sales	24,914,586.82	Yes
Total Cost	14,456,871.39	Yes
Total Profit	10,457,715.43	Yes
Total Quantity	84,174	Yes
Total Orders	25,164	Yes
Total Customers	17,416	Yes
Average Sales (per line)	444.54	Yes
Average Order Value	990.09	Yes
Average Product Price	714.44	Yes
Profit Margin %	41.97	Yes

Query 2: Top Ten Products by Sales

```
EVALUATE
```

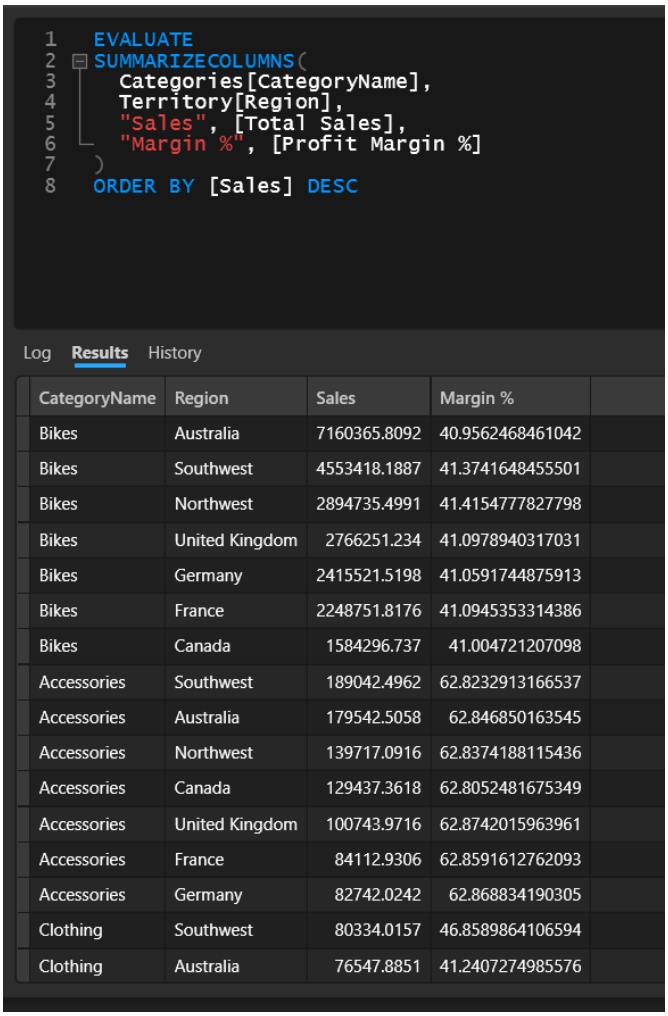
```
TOPN(10, SUMMARIZECOLUMNS(Products[ProductName], "Sales", [Total Sales],
"Profit", [Total Profit]),
  [Sales], DESC)
ORDER BY [Sales] DESC
```



ProductName	Sales	Profit
Mountain-200 Black, 46	1241753.5092	571632.6492
Mountain-200 Black, 42	1233557.1164	567859.4964
Mountain-200 Silver, 38	1213851.8856	558788.3282
Mountain-200 Silver, 46	1182780.5916	544484.8727
Mountain-200 Black, 38	1165936.8758	536730.9858

Query 3: Sales and Profit Margin by Category and Region

```
EVALUATE
SUMMARIZECOLUMNS(Categories[CategoryName], Territories[Region],
  "Sales", [Total Sales], "Margin %", [Profit Margin %])
ORDER BY [Sales] DESC
```

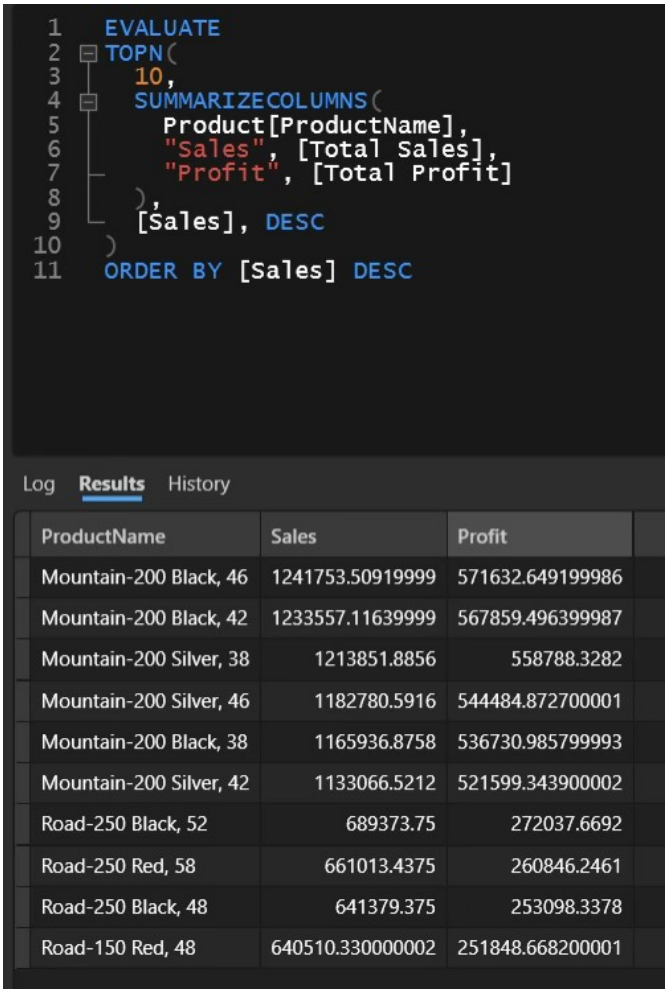


CategoryName	Region	Sales	Margin %
Bikes	Australia	7160365.8092	40.9562468461042
Bikes	Southwest	4553418.1887	41.3741648455501
Bikes	Northwest	2894735.4991	41.4154777827798
Bikes	United Kingdom	2766251.234	41.0978940317031
Bikes	Germany	2415521.5198	41.0591744875913
Bikes	France	2248751.8176	41.0945353314386
Bikes	Canada	1584296.737	41.004721207098
Accessories	Southwest	189042.4962	62.8232913166537
Accessories	Australia	179542.5058	62.846850163545
Accessories	Northwest	139717.0916	62.8374188115436
Accessories	Canada	129437.3618	62.8052481675349
Accessories	United Kingdom	100743.9716	62.8742015963961
Accessories	France	84112.9306	62.8591612762093
Accessories	Germany	82742.0242	62.868834190305
Clothing	Southwest	80334.0157	46.8589864106594
Clothing	Australia	76547.8851	41.2407274985576

Query 4: Top Ten Products, Complete Result Set

Query 2 displayed only the first rows of the top-ten ranking. This query returns the complete ranking. SUMMARIZECOLUMNS groups the data by product name and evaluates Total Sales and Total Profit for each group, TOPN(10, …) retains the ten products with the highest sales, and ORDER BY presents them in descending order.

```
EVALUATE
TOPN(
  10,
  SUMMARIZECOLUMNS(
    Product[ProductName],
    "Sales", [Total Sales],
    "Profit", [Total Profit]
  ),
  [Sales], DESC
)
ORDER BY [Sales] DESC
```



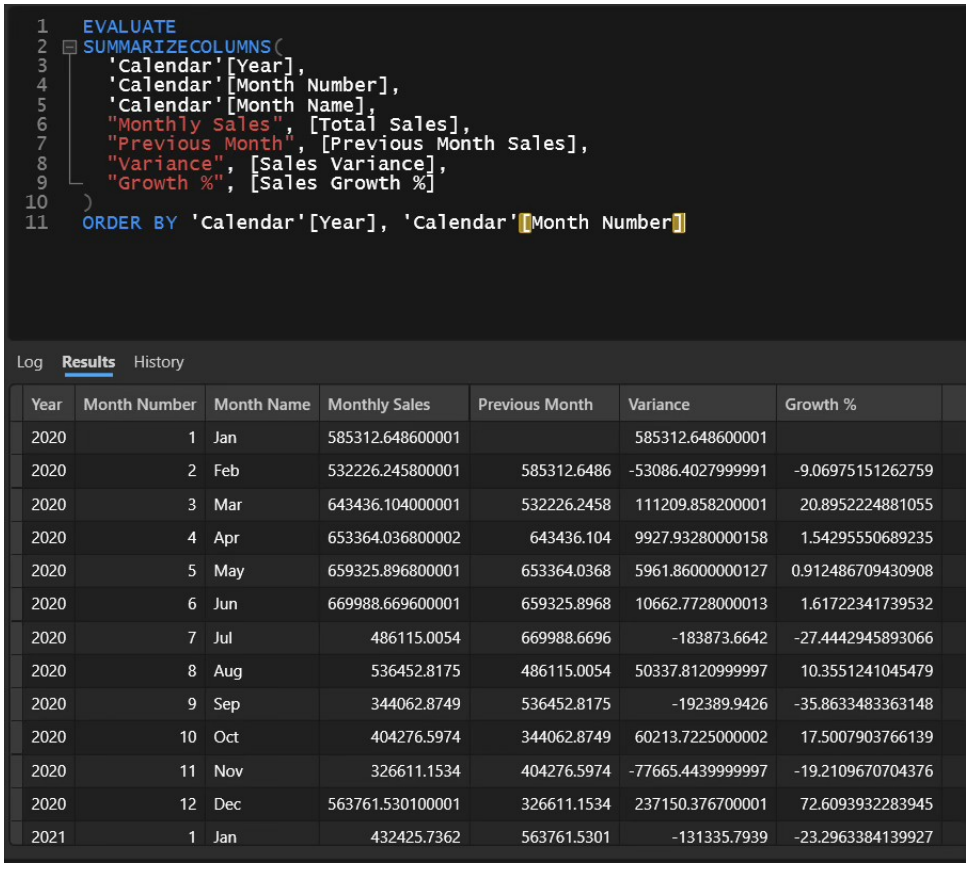
ProductName	Sales	Profit
Mountain-200 Black, 46	1241753.50919999	571632.649199986
Mountain-200 Black, 42	1233557.11639999	567859.496399987
Mountain-200 Silver, 38	1213851.8856	558788.3282
Mountain-200 Silver, 46	1182780.5916	544484.872700001
Mountain-200 Black, 38	1165936.8758	536730.985799993
Mountain-200 Silver, 42	1133066.5212	521599.343900002
Road-250 Black, 52	689373.75	272037.6692
Road-250 Red, 58	661013.4375	260846.2461
Road-250 Black, 48	641379.375	253098.3378
Road-150 Red, 48	640510.330000002	251848.668200001

Observations. The six Mountain-200 variants occupy ranks one to six, with sales between about $1.13M and $1.24M each and combined sales of about $7.17M (roughly 28.8% of total sales). The Road-250 and Road-150 variants take ranks seven to ten, with sales of about $0.64M to $0.69M each. Every Mountain-200 variant earns a profit of about 46.0% of its sales, compared with about 39.3% to 39.5% for the Road variants. The ten products together generate about $9.80M, or roughly 39% of total sales.

Query 5: Monthly Sales, Previous Month, Variance and Growth

This query shows the output of the time-based measures defined in Section 5. It groups the data by year and month and evaluates four measures for each month: current sales, the sales of the preceding month, the difference between the two (Sales Variance) and the percentage change (Sales Growth %). Month Number is included so that months sort chronologically instead of alphabetically.

```
EVALUATE
SUMMARIZECOLUMNS(
  'Calendar'[Year],
  'Calendar'[Month Number],
  'Calendar'[Month Name],
  "Monthly Sales", [Total Sales],
  "Previous Month", [Previous Month Sales],
  "Variance", [Sales Variance],
  "Growth %", [Sales Growth %]
)
ORDER BY 'Calendar'[Year], 'Calendar'[Month Number]
```



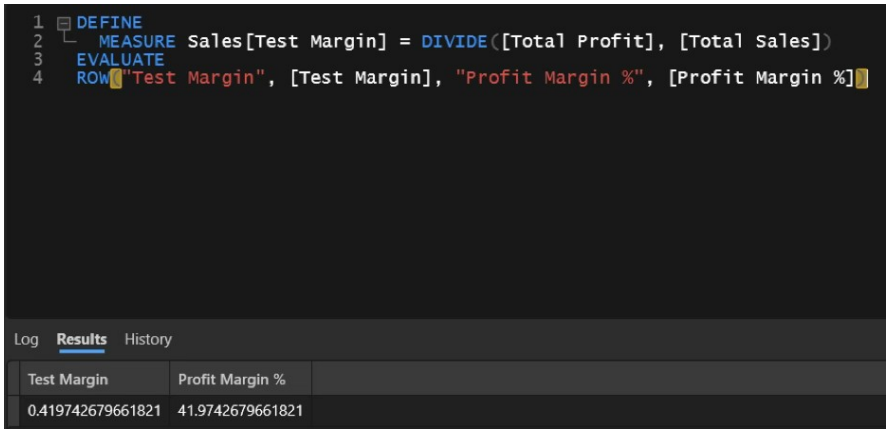
Year	Month Number	Month Name	Monthly Sales	Previous Month	Variance	Growth %
2020	1	Jan	585312.648600001		585312.648600001	
2020	2	Feb	532226.245800001	585312.6486	-53086.4027999991	-9.06975151262759
2020	3	Mar	643436.104000001	532226.2458	111209.858200001	20.8952224881055
2020	4	Apr	653364.036800002	643436.104	9927.93280000158	1.54295550689235
2020	5	May	659325.896800001	653364.0368	5961.86000000127	0.912486709430908
2020	6	Jun	669988.669600001	659325.8968	10662.7728000013	1.61722341739532
2020	7	Jul	486115.0054	669988.6696	-183873.6642	-27.4442945893066
2020	8	Aug	536452.8175	486115.0054	50337.8120999997	10.3551241045479
2020	9	Sep	344062.8749	536452.8175	-192389.9426	-35.8633483363148
2020	10	Oct	404276.5974	344062.8749	60213.7225000002	17.5007903766139
2020	11	Nov	326611.1534	404276.5974	-77665.4439999997	-19.2109670704376
2020	12	Dec	563761.530100001	326611.1534	237150.376700001	72.6093932283945
2021	1	Jan	432425.7362	563761.5301	-131335.7939	-23.2963384139927

Observations. January 2020 is the first month in the data, so it has no previous-month value or growth percentage, and its variance equals its own sales of about $585K. Sales then rose from February to June 2020, reaching about $670K in June. They fell sharply in July 2020 (−27.4%) and again in September 2020 (−35.9%), then recovered to about $564K in December 2020 (+72.6%). January 2021 dipped by about 23.3% to roughly $432K. The screenshot shows January 2020 to January 2021 only; the remaining months are available in the same result set.

Query 6: Testing a Measure with DEFINE MEASURE

DAX Studio allows a measure to be defined for the duration of a single query using DEFINE MEASURE, so a formula can be tested without changing the Power BI model. Here, a temporary measure named Test Margin calculates profit divided by sales as a plain ratio, and it is returned alongside the model's own Profit Margin % measure.

```
DEFINE
  MEASURE Sales[Test Margin] = DIVIDE([Total Profit], [Total Sales])
EVALUATE
ROW("Test Margin", [Test Margin], "Profit Margin %", [Profit Margin %])
```

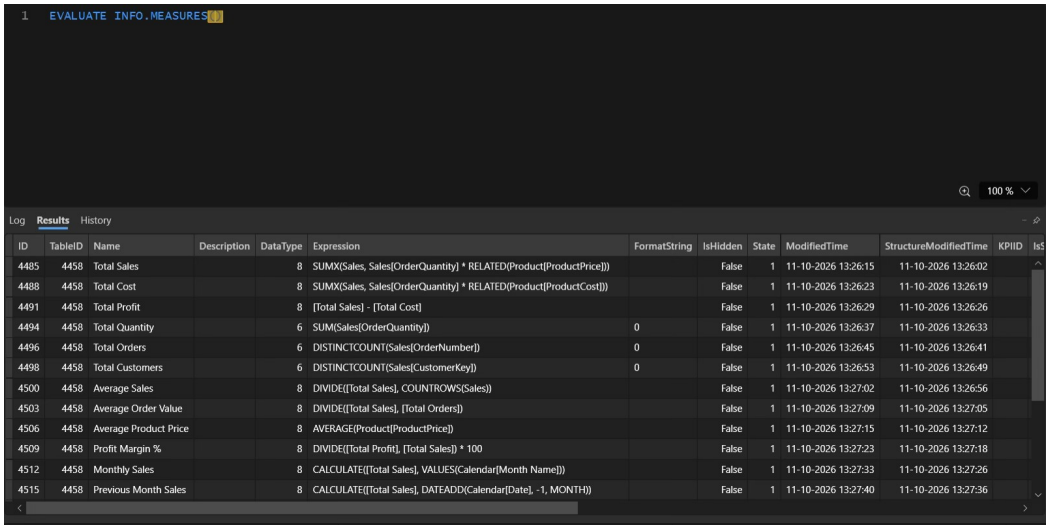


Observations. Test Margin returns 0.41974…, and Profit Margin % returns 41.9742…. The two differ only by the factor of 100 built into Profit Margin %, so the model's formula is consistent with the independent calculation. Both agree with the overall margin of 41.97% reported in Query 1.

Query 7: Auditing the Measures in the Model

INFO.MEASURES() is a DAX function that returns metadata for every measure in the model. It was used to review the definitions stored in the model and to confirm which measures exist.

```
EVALUATE INFO.MEASURES()
```

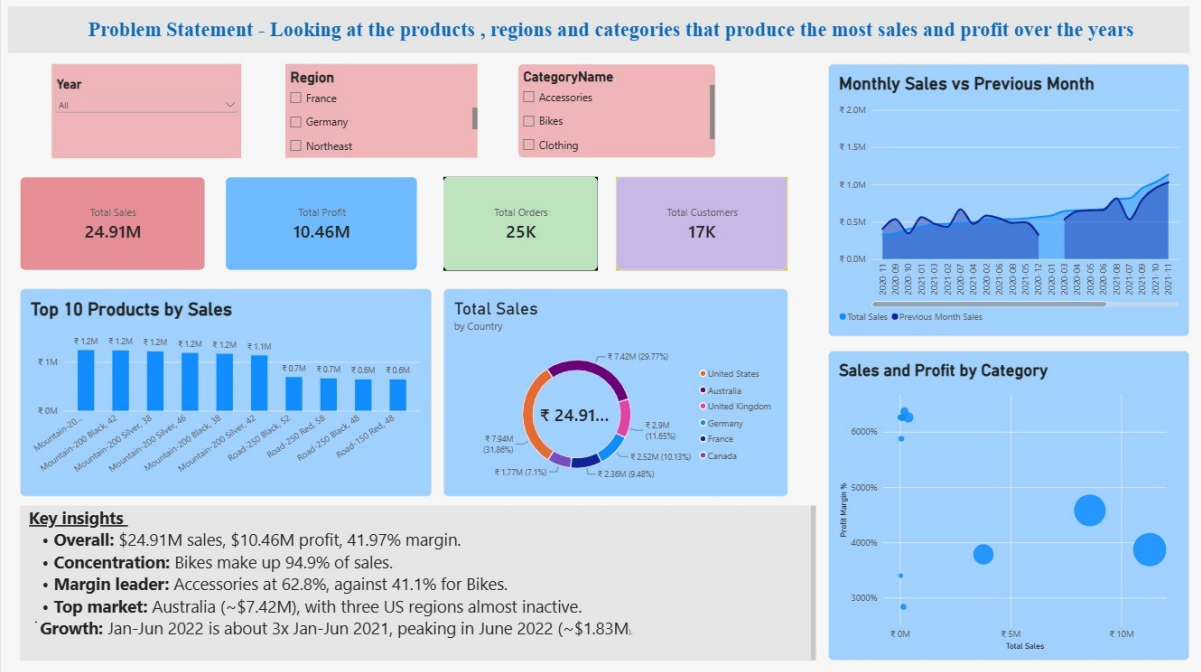


ID	TableID	Name	Description	DataType	Expression	FormatString	IsHidden	State	ModifiedTime	StructureModifiedTime	KPIID	IsS
4485	4458	Total Sales		8	SUMX(Sales, Sales[OrderQuantity] * RELATED(Product[ProductPrice]))		False	1	11-10-2026 13:26:15	11-10-2026 13:26:02		
4488	4458	Total Cost		8	SUMX(Sales, Sales[OrderQuantity] * RELATED(Product[ProductCost]))		False	1	11-10-2026 13:26:23	11-10-2026 13:26:19		
4491	4458	Total Profit		8	[Total Sales] - [Total Cost]		False	1	11-10-2026 13:26:29	11-10-2026 13:26:26		
4494	4458	Total Quantity		6	SUM(Sales[OrderQuantity])	0	False	1	11-10-2026 13:26:37	11-10-2026 13:26:33		
4496	4458	Total Orders		6	DISTINCTCOUNT(Sales[OrderNumber])	0	False	1	11-10-2026 13:26:45	11-10-2026 13:26:41		
4498	4458	Total Customers		6	DISTINCTCOUNT(Sales[CustomerKey])	0	False	1	11-10-2026 13:26:53	11-10-2026 13:26:49		
4500	4458	Average Sales		8	DIVIDE([Total Sales], COUNTROWS(Sales))		False	1	11-10-2026 13:27:02	11-10-2026 13:26:56		
4503	4458	Average Order Value		8	DIVIDE([Total Sales], [Total Orders])		False	1	11-10-2026 13:27:09	11-10-2026 13:27:05		
4506	4458	Average Product Price		8	AVERAGE(Product[ProductPrice])		False	1	11-10-2026 13:27:15	11-10-2026 13:27:12		
4509	4458	Profit Margin %		8	DIVIDE([Total Profit], [Total Sales]) * 100		False	1	11-10-2026 13:27:23	11-10-2026 13:27:18		
4512	4458	Monthly Sales		8	CALCULATE([Total Sales], VALUES(Calendar[Month Name]))		False	1	11-10-2026 13:27:33	11-10-2026 13:27:26		
4515	4458	Previous Month Sales		8	CALCULATE([Total Sales], DATEADD(Calendar[Date], -1, MONTH))		False	1	11-10-2026 13:27:40	11-10-2026 13:27:36		

Observations. Each row describes one measure: its ID, the table that holds it (TableID), its Name, DataType, the DAX Expression, FormatString, whether it is hidden (IsHidden), State, and the ModifiedTime and StructureModifiedTime timestamps. The screenshot shows the first thirteen measures, from Total Sales to Previous Month Sales. Scrolling the result grid shows the remaining measures.

7. Interactive Power BI Dashboard (Task 5)

The dashboard comprises four KPI cards (Total Sales, Total Profit, Total Orders and Total Customers), slicers for Year, Region and CategoryName, four visualizations and a key-insights panel. Selecting a value in any slicer dynamically updates every card and chart on the page.



Visual	Fields	Purpose
Area / line chart	X: Year-Month; Y: Total Sales, Previous Month Sales	To examine the monthly trend against the previous month
Column chart	X: ProductName (Top 10); Y: Total Sales	To identify the best-selling products
Donut chart	Legend: Country; Values: Total Sales	To identify the countries contributing most to sales
Scatter chart	X: Total Sales; Y: Profit Margin %	To compare sales volume with profitability by category

8. Analysis and Interpretation of Results (Task 6)

- **Overall performance:** The business recorded sales of approximately $24.91M and profit of $10.46M, giving a profit margin of 41.97%. These figures arise from 25,164 orders placed by 17,416 customers, with an average order value of roughly $990.
- **Top-selling products:** The highest-ranked products are all Mountain-200 bikes. Mountain-200 Black 46 leads with about $1.24M, followed by Black 42 (about $1.23M) and Silver 38 (about $1.21M).
- **Category performance:** Bikes generate about $23.64M, or 94.9% of total sales. Accessories contribute roughly $0.91M and Clothing $0.37M. No sales are recorded for Components in this dataset.
- **Regional performance:** Australia ranks first with about $7.42M, close to 30% of total sales, followed by Southwest (about $4.82M), Northwest (about $3.10M) and the United Kingdom (about $2.90M). Three US regions, namely Central, Northeast and Southeast, record negligible sales of under $12K each.
- **Sales trend:** Annual sales increased from $6.40M in 2020 to $9.32M in 2021. The 2022 data covers January to June only, yet sales had already reached $9.19M, approximately three times the figure for the same six months of 2021. Monthly sales show a consistent upward trend, with June 2022 the strongest month recorded at about $1.83M, an increase of around 3.3% over May.
- **Profitability:** Bikes account for the highest sales but have the lowest profit margin (41.1%). Accessories yield the highest margin per dollar sold (62.8%), while Clothing occupies an intermediate position (44.3%). High revenue therefore does not necessarily imply the highest margin.

Category	Sales ($)	Profit margin	Share of sales
Bikes	23.64M	41.1%	94.9%
Accessories	0.91M	62.8%	3.6%
Clothing	0.37M	44.3%	1.5%
Components	No sales	-	-

9. Conclusion

The analysis shows that the sales and profit of Adventure Works are driven overwhelmingly by the Bikes category, led by the Mountain-200 range, and by a small number of markets, with Australia ranking first. Accessories deliver the highest profit margin (62.8%) but account for only 3.6% of sales, which suggests that expanding this category is the most direct means of improving overall margin. Sales exhibit a steady month-on-month increase, and sales for the first half of 2022 are already close to the full-year total for 2021. The near-inactivity of three US regions indicates untapped potential, while the absence of sales for Components warrants verification of the source data.

The twenty-five DAX measures were validated against the raw data, and the dashboard enables these patterns to be explored by year, region and category.